

C - Malloc with strings

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

int main() {
    char a[] = "hello";
    char b[] = "world";

    printf("a = %s\n", a);
    printf("b = %s\n", b);

    // Determine the size needed for the temporary buffer
    size_t size_a = strlen(a) + 1; // +1 for the null
terminator
    size_t size_b = strlen(b) + 1; // +1 for the null
terminator

    // Allocate memory for the larger of the two strings
    size_t temp_size = size_a > size_b ? size_a : size_b;
    char *temp = (char *)malloc(temp_size);

    // Check if memory allocation was successful
    if (temp == NULL) {
        fprintf(stderr, "Memory allocation failed!\n");
        return 1;
    }

    // Swapping with a dynamically allocated temporary buffer
    strcpy(temp, a); // Copy 'hello' into temp
    strcpy(a, b);    // Copy 'world' into a
    strcpy(b, temp); // Copy 'hello' from temp into b

    // Free the temporary buffer
```

```
    free(temp);

    printf("After swap:\n");
    printf("a = %s\n", a);
    printf("b = %s\n", b);

    return 0;
}
```

- We use `strlen` to find the length of each string and add 1 to account for the null terminator.
- We then allocate a buffer of the appropriate size using `malloc`.
- After using the temporary buffer to swap the strings, we free the allocated memory using `free`.